

Index

Note: Page numbers followed by ‘f’ indicate figures, ‘t’ indicates tables, and ‘b’ indicates boxes.

A

A330 airbus model, 413b–414b
A/B testing, 371, 438–441
ABC News Nightline 1999, 242b–243b, 252
Abstract work activity, 214–215
 top-down design approach, 215–221
 characteristics, 219–220
 disruptive design, 220–221
 in architecture, 220b–221b
Abstraction, 35–36, 507f
 in ideation, 237
Acceptance test, 510
Action-object approach, 156b
Active entities, social model, 169
Activity theory, roots in, 105
Activity-based interaction, 5–7
 and broader ecology, 103
 multiple devices, 6
 one device, 5–6
 user’s ecology, 6
Aesthetics, product design, 11
Affinity diagrams, 34–35, 119–120
Affordances
 acknowledgement of source, 527
 cognitive. *See* Cognitive affordances
 concept of, 527
 design, 558–562
 affordance design checklist, 562b
 cognitive affordance considerations, 560
 emotional affordance considerations, 561–562
 functional affordance considerations, 560
 physical affordance considerations,
 560–561, 560f
 roles, 558–559

Affordances (*Continued*)

 sensory affordance considerations, 561
 UX design checklist, 559–562
early human-computer interaction and UX, 528
emotional affordance, 557–558
functional affordance, 556–557, 557f
history of, 565–566
importance of, 531
physical. *See* Physical affordances
sensory. *See* Sensory affordance
shared cultural conventions, 528–531
types, 532t, 533f
United States *vs.* Europe, 529
user-created affordances, 562–564, 562b–564b, 563f,
 565f
UX design, 531–532
 issues, 531
Agile lifecycle process, 58
Agile SE sprints, 64
Agile software engineering (SE), 501–503
 agile UX, synthesized approach to, 514–523
 customer prioritize user stories, 517
 late-funnel mini-UX process checklist, 519t
 planning, 515–517, 515f
 small upfront analysis and design, 516–517
 sprints, 517–519, 518f
 truth about user stories, 517
 anticipate problems, 513–514
 overview, 514
 UX, 513–514
 basics of, 503–506
 avoiding big design upfront, 506
 characteristics of, 505
 goals, 503–504

Agile software engineering (SE) (*Continued*)

- principles of, 503–504
 - silos operating, 505
 - waterfall method, contrasting with, 504–506
 - challenges of, 511
 - example, 502b
 - lifecycle aspects, 506–511, 506f
 - abstraction of, 507f
 - controlling scope, 509
 - customer stories, 507
 - managing customer stories and development tasks, 509
 - planning, 507–509, 507f
 - sprints. *See* Sprints
 - story-based planning, 508, 508f
 - two agile workflows, synchronizing, 520–523
 - communication during synchronization, 523
 - continuous delivery, 522
 - dovetailed work activities, 520–521, 520f
 - planning across iterations, 522–523
 - regression testing, 522
 - value of early delivery, 521–522
 - UX side, needed on, 512
- Agile UX funnel model, 182
- Agile UX method
- funnel model, 61–67, 63f
 - design creation sub lifecycle, 207, 207f
 - early-funnel activities, 65–67, 66b
 - late-funnel activities, 63–65, 64f
 - new model needed for, 62
 - scope, 63
 - speed and rigor in, 63, 64f
 - lifecycle process, 39, 58–61
 - agility without chunking, 60
 - chunking, 59–60, 59f
 - mapping project parameters, 40–41, 41f
 - reasons for, 61
 - scope and elicit quick feedback, 59–60
 - software engineering, 60–61
 - spectrum, bookends on, 59
 - user-facing prototypes, 60–61
 - users, interest in, 61
 - managing change challenges, 55

Agile UX method (*Continued*)

- lifecycle process, 56
 - timely feedback, 56
 - waterfall model, software system development, 56–58, 57f
- Agile UX, synthesized approach to, 514–523
- customer prioritize user stories, 517
 - late-funnel mini-UX process checklist, 519t
 - planning, 515–517, 515f
 - small upfront analysis and design, 516–517
 - sprints, 517–519, 518f
 - truth about user stories, 517
- Alpha/beta testing and field surveys, 437
- Alternative data sources, 93
- Ambiguity in scope, 198–200
- Analytic formative evaluation, 372–373
- Analytic UX evaluation, data collection, 443
- adding, 443–445, 444f
 - characteristics, 444
 - criticism of, 445
 - design walk-throughs, 446–449
 - artifacts involved in, 447
 - click-through prototypes value, 448–449
 - conduct, 448–449
 - expected feedback, 447
 - goals for, 447
 - parameters for, 447
 - prepare for, 447–448
 - emotional impact inspection, 457–459, 458b–459b
 - focus groups, 445–446
 - reality of, 446
 - UX inspection
 - broad range of, 450
 - co-discovery approach, 456–457
 - defined, 449
 - heuristic evaluation, 451–456
 - insight and experience, 456
 - inspectors needed for, 450–451
 - kind of expertise, 450
 - practical approach to, 456–457
 - rich and comprehensive usage-oriented view, 457
 - tasks and design driven, 457
- Andres, C., 184

Animated prototypes, 360–361
 Ann, E., 20–21
 Anthropomorphism, 666–669
 avoiding, 666–667
 case in favor of, 667–669
 Apple Music, 274b–275b, 295–296
 Arnowitz, J., 62
 Artifact model, 131, 189, 231
 auto repair shop, 147b
 construction, 145–147
 defined, 143–147
 restaurant flow model, 146f, 146b–147b
 Artifacts, 447
 Association for Computing Machinery (ACM), 20–21
 AttrakDiff questionnaire, 427
 Audio characteristics, 331–332
 Auditory sensory issues, 555, 555b
 Automatic UX evaluation, 438
 Avoid jargon, 497

B

Baskinger, M., 245, 254
 Baty, S., 225
 Beale, R., 287–288
 Beck, K., 184
 Benchmark tasks, 386–388
 descriptions, 400
 Beu, A., 11
 Beyer, H., 139, 159, 196, 211
 Biases, 211–213
 Black-on-black in stereo, 550–552, 552f
 Blackberry smartphone, 211–212
 Bodker, S., 71
 Boehm, B. W., 202
 Boling, E., 235
 Borchers, J., 298
 Bottom-up UX design approach
 biases and constraints, roles, 211–213
 existing work practice, designing for, 209–213
 human-centered/user-centered design,
 210–211
 innovative possibilities, 213
 Brainstorming, 38
 Branding, emotional impact, 14–15
 Buchenau, M., 361
 Burmester, M., 11
 Business case, for UX, 19–20
 Buur, J., 71
 Buxton, B., 13, 200, 244–245, 252

C

Calibration feedback, 475
 Car wash workflow, 97b
 Card sorting, 34, 119
 Carroll, J. M., 258, 269
 Catalysts, ideation, 227
 eureka moment, 235
 Chunking, 59–60, 59f, 584–585, 584b–585b
 agility without, 60
 Click-through prototypes value, 448–449
 Co-discovery think-aloud techniques, 426
 Codiscovery evaluation, 402–403
 Cognitive affordances, 528–531, 534–540
 cognitive support information, 534–535, 535b
 content and meaning design issues, 537–540
 cultural conventions, 536b
 definition, 534
 false cognitive affordances, 538–540, 538b–539b
 mail icon as, 535b
 opening doors devices, 536
 shared conventions, 535–536
 task completion errors, 538, 538f
 user, 537–538
 Cognitive load, 585–586
 Commercial tools for domain, 90
 Communication during synchronization, 523
 Completeness, 43
 Complexity influences
 complex interaction, complex work domain, 46–47
 complex interaction, simple work domain, 47
 domain complexity, 45
 interaction complexity, 45
 simple interaction
 complex work domain, 47–48
 simple work domain, 45–46
 system complexity space, 44

Complexity influences (*Continued*)

- gradations within, 48
- quadrants, 45–48

Computer graphics, origin of, 23

Computer mouse, first, 22f

Computing, evolution of, 4

Concept mapping, 34

Concept/mind mapping, 119

Conceptual design, 263–265

- as connection between designer and user mental models, 256–265
- as mapping, 259–262
- for ecology, 259–262, 284
- for emotional impacts, 268
- interaction, 259–262, 267–268
- Johnson and Henderson's rule, 265
- metaphors, 269–271
- patterns, 269
- UX design starts with, 265–271
- work practice ecology, 266–267

Conference on Human Factors in Computer Systems (CHI), 20–21

Constantine, L. L., 155, 165–166, 166b

Constraints, 211–213

Consumer products, 53–54

Continuous delivery, 522

Controlling scope, 509

Convergence, 465

Cooper's in-flight entertainment system, 230b

Cooper, A., 16, 133–134, 136–137, 229, 230b

Cooperative user-system task performance, 576–578

- intention shifts, 577–578
- path variations, 576–577, 577f
- primary tasks, 576
- secondary tasks, 577–578
- stacking, 577–578, 579f

Corporate culture, emotional impact, 14–15

Cost-benefit analysis, 20

Cost-importance analysis, 471–484

- calibration feedback, 475
- cost to fix, 474–475
- cost values for problem groups, 475
- cumulative cost, 479

Cost-importance analysis (*Continued*)

- drawing conclusion, 481, 482t
- emotional impact problems priorities, 483–484
- importance rating adjustments, 473
- importance to fix, 472–473
- line of affordability, 479–481, 480t, 483
- multiple problem solutions, 483
- priority rankings, 477–479, 479f
- priority ratio, 475–477, 476b, 477t
- problem, 471–472, 472t
- rapid cost-importance analysis, 484
- solutions, 474, 474t, 474b
- tie-breakers, 483

Cost justification, 20

Counting user errors, 415

Crafting benchmark task descriptions, 387–388

Critical incident technique, 422–424

Criticism of discount methods, 378

Critiquing, 250–251, 251f

- rules of engagement, 251–252

Cross, N., 236

CSAT, 371–372

Cultural conventions, 536b

Cumulative cost, 479

Customer prioritize user stories, 517

Customer service and support, 6b

Customer stories, 507

D

Dagstuhl framework, interaction, 8

Dancing bear software, 16

Data analysis, 30, 461

- consolidate congruent data notes, 469
- cost-importance analysis, 471–484
 - calibration feedback, 475
 - cost to fix, 474–475
 - cost values for problem groups, 475
 - cumulative cost, 479
 - drawing conclusion, 481, 482t
 - emotional impact problems priorities, 483–484
 - importance rating adjustments, 473
 - importance to fix, 472–473
 - line of affordability, 479–481, 480t, 483

Data analysis (*Continued*)

- multiple problem solutions, 483
- priority rankings, 477–479, 479f
- priority ratio, 475–477, 476b, 477t
- problem, 471–472, 472t
- rapid cost-importance analysis, 484
- solutions, 474, 474t, 474b
- tie-breakers, 483
- elemental data notes, 467
- feedback to process, 484–485
- group-related UX problem descriptions, 469–470
- onion layer effect, 485–486, 485b–486b
- qualitative data
 - analysis preparation steps, 466
 - analysis steps, 466–471, 467f
 - overview, 465
- quantitative data, 461–465
 - big decision, 464–465
 - convergence, 465
 - line up your quantitative ducks, 463–464
 - meet target column, 463–464
 - observed results, 463
 - simple descriptive statistics, 462–463
 - subjective quantitative questionnaire data, 463
- Ticket Kiosk System, 470b
- UX research analysis tools, 471

Data collection emotional impact

- AttrakDiff questionnaire, 427
- direct observation, 426
- self-reporting technique questionnaires, 427
- use think-aloud technique, 427

Data gathering, 30, 83–84, 98

Data loss risk, 129–131

Data modeling, usage research, 30

- artifact model, 131
 - construction, 145–147
 - defined, 143–147
- flow model, 139–143
 - central importance, 139
 - customer journey map, 143
 - defined, 139
 - for product/system, 142b–143b
 - making of, 139–143

Data modeling, usage research (*Continued*)

- hybrid model, 171
- information architecture model,
 - 150–151
- kinds of, 131
- physical work environment model
 - environmental factor, 148
 - hardware design, 147
- social model
 - active entity, 169
 - concerns, 169–171
 - for product/system, 171b
 - influences, 169
 - issues with, 169–171
 - shared workplace culture, 168
 - simplified approach, 168–169
- task sequence model, 159b, 167b
 - beyond linear model, 163–164
 - components, 160
 - defined, 158–168
 - essential use case, 165–167
 - for product/system, 159b
 - menu planning application, 159b
 - state diagrams, 167
 - step-by-step description, 159–163
 - task interaction model, 161b–163b
- user personas
 - creation, 138
 - data extraction, 137–138
 - defined, 137
 - early sketch, 138b
 - Ticket Kiosk System, 138b
- work role model, 132–136
 - defined, 132–133
 - for product/system, 134b
 - mediated work role, 133–134
 - subroles, 133
 - user class, 134–136
- wrap up
 - barrier summaries across all
 - model, 173
 - UX design studio, 173

Data sources, 93–94

- Deeper emotions, 12–13
- Depth and breadth, prototyping, 340f
 - examples of kinds, 342–343
 - horizontal prototype, 342
 - horizontal prototypes, 340
 - local prototypes, 341–343
 - T prototypes, 341–343
 - vertical prototypes, 340–342
- Design, 27–31, 195f
 - activity-based, 5
 - defined, 193–195
 - efficiency, 627–628
 - flexibility, 627
 - human needs, satisfying, 200–201
 - interdisciplinary nature, 198
 - iteration
 - conceptual, 204
 - detailed, 206
 - generative, 202–203
 - intermediate, 204–206
 - lifecycle
 - process, 27
 - template, 28, 28f–29f
 - as noun, 195
 - patterns, 239, 298
 - in conceptual design, 269
 - purpose of, 200–201
 - scenario, 158–159
 - SE implementation, 206
 - solutions lifecycle activity, 194f, 199f, 224f, 296f
 - macro view, 203f
 - synthesis, 225
 - thinking, 19
 - universality, 196–197
 - as verb, 195
 - walk-throughs, 446–449
 - artifacts involved in, 447
 - click-through prototypes value, 448–449
 - conduct, 448–449
 - expected feedback, 447
 - goals for, 447
 - parameters for, 447
 - prepare for, 447–448
- Design situations, 40
 - consumer products, 53–54
 - enterprise systems, 54
 - lifecycle activities, methods, and techniques, 51–52
 - rigor, 43–51
 - complexity influences, 44–48
 - development stages, 50
 - domain familiarity influences, 49
 - project resources influence, 51
 - risk aversion influences, 49–50
 - UX team experience, 51
 - scope of delivery, 52–53
- Design techniques as life skills, UX, 30–31
 - abstraction, 35–36
 - brainstorming, 38
 - framing and reframing, 36
 - immersion, 35
 - modeling, 37–38
 - note taking, 32
 - observation, 31–32
 - prototyping for envisioning, 38
 - reasoning and deduction, 36
 - sketching, 32
 - storytelling, 38
 - synthesis, 37
- Designers, 224
 - mental model, 257f, 258, 267
 - workflow, 266f
- Detailed design iteration, 206
- Diary-based self-reporting by users, 428–429
- Direct observation, 426
- Disappearing technology, 4
- Discount methods, criticism of, 378
- Domain familiarity influences, 49
- Donohue, J., 227
- Dorst, K., 236
- Dourish, P., 253–254
- Dovetailed work activities, 520–521, 520f
- Draper, S. W., 210
- Dual experts, 93

E

- Early funnel data gathering, 95
- Early-funnel activities, 65–67, 66b
- Ecological design, 273
 - conceptual design, 285b–287b
 - creation of, 283–287
 - designing for, 273–279
 - information architecture, 275–276
 - interaction channels, 276–277
 - smart shopping application, 289–294
 - storyboards, 279–283
 - usage context, 275
 - user behavior, 287–289
 - user frustration, 278b–279b
- Ecological level of the pyramid of needs, 373–374
- Ecological perspectives, 201
- Elemental data notes, 467
 - user persona model, 137
- Embodied interaction, 240–241
- Emotional affordance, 557–558
- Emotional impact design, 10–15, 189, 323
 - branding, 14–15, 15b
 - conceptual design, 332b
 - example, 327b–328b
 - metaphors, 326
 - mood boards, 327–328, 327f
 - coolness and wow in, 14
 - corporate culture, 14–15
 - deeper emotions, 12–13
 - emotional needs, 324–325
 - to satisfy, 325–326, 326f
 - users crave distinctiveness, 324–325
 - fun interaction at work, 12
 - importance of, 14b
 - intermediate design
 - audio characteristics, 331–332
 - motion styles and physics, 329–331, 330f
 - social and psychological aspects, 332
 - tone of language, 331
 - visual language and vocabulary, 328–329, 329b
 - joy, excitement, and fun, 13, 13f–14f
 - market differentiator, 323
 - marketing, 14–15
 - Emotional impact design (*Continued*)
 - passion and, 15b
 - production, 332, 333f
 - segue, 333
 - Emotional impact inspection, 457–459, 458b–459b
 - Emotional level of the pyramid of needs, 374
 - Emotional needs, 324–325
 - to satisfy, 325–326, 326f
 - users crave distinctiveness, 324–325
 - Emotional perspectives, 201
 - Empirical data collection sessions procedures, 429–433
 - facilitator interacts with participants, 430–431
 - facilitator keeps the participants at ease, 431–433
 - participants preliminaries, 429
 - process and expectations, 429
 - product/system, 431b–433b
 - prototype evaluating, 429–430
 - session begins, 430
 - Empirical formative UX evaluation, 372
 - Empirical method, 412–413
 - Empirical UX evaluation, data collection, 411, 412f
 - ecological validity, 413–414, 413b–414b
 - emotional impact
 - AttrakDiff questionnaire, 427
 - direct observation, 426
 - self-reporting technique questionnaires, 427
 - use think-aloud technique, 427
 - empirical data collection sessions, procedures for, 429–433
 - facilitator interacts with participants, 430–431
 - facilitator keeps the participants at ease, 431–433
 - participants preliminaries, 429
 - process and expectations, 429
 - product/system, 431b–433b
 - prototype evaluating, 429–430
 - session begins, 430
 - empirical method, 412–413
 - meaningfulness, 427–429
 - diary-based self-reporting by users, 428–429
 - evaluation requires long-term studies, 428
 - self-reporting, importance of, 428
 - qualitative
 - co-discovery think-aloud techniques, 426

Empirical UX evaluation, data collection (*Continued*)

- critical incident technique, 422–424
- user think-aloud data collection techniques, 425–426

quantitative

- counting user errors, 415
- objective data collection, 415
- questionnaires, 416–421
- slip is not a user error, 415
- subjective data collection, 416–421
- timing task performance, 415

rapid empirical UX evaluation methods, 433–436

- introduction to, 433
- quasi-empirical UX evaluation, 433–434
- rapid iterative testing and evaluation (RITE), 434–436
- variations, 436

specialized UX evaluation methods, 437–441

- A/B comparison testing, 438–441
- alpha and beta testing and field surveys, 437
- automatic UX evaluation, 438
- post-deployment field testing, 437
- remote UX evaluation, 437–438

treatment of, 413

Empirical UX evaluation, preparation, 397

- evaluation rigor, 399
- evaluation scope, 399
- goals for, 399
- plan, 397–398
- prepare for the session
 - final pilot testing, 409
 - informed consent, 406–407
 - lab and equipment, 403, 404f
 - non-disclosure agreement (NDA), 408
 - other paperwork, 408–409
 - questionnaires and surveys, 408–409
 - session parameters, 404–406, 405f
- preparing for, 398f
- recruit participants, 401–403
 - budget and schedule for, 401
 - codiscovery evaluation, 402–403
 - identify right kinds of participants, 401
 - incentives and remuneration, 402

Empirical UX evaluation, preparation (*Continued*)

- recruiting methods and screening, 402
- right number of participants, 402
- subsequent iterations, select participants for, 403
- select team roles, 399–400
 - facilitator, 400
 - participation and buy-in, 399
 - prototype executor, 400
 - qualitative data collectors, 400
 - quantitative data collectors, 400
- subsequent iterations, select participants for, 403
- user tasks and appropriate survey instruments,
 - effective range of
 - benchmark task descriptions, 400
 - exploratory free use tasks, 401
 - other kinds of task descriptions, 401
 - unmeasured tasks, 401
 - user-defined tasks, 401

Engelbart, D., 22

Enterprise systems, 47, 54

Ergonomics, 21

Essential use case task sequence models, 165–167

Ethnographic-based approach, 106

Ethnography, roots in, 105–106

Evaluation rigor, 399

Evaluation scope, 399

Experience prototyping, 361–362

Exploratory free use tasks, 401

Externalizing insights, 39

F

Facilitator, 400

- interacts with participants, 430–431
- participants at ease, 431–433

False cognitive affordances, 538–540, 538b–539b

Feed-forward

- existence of, 593–595, 593f, 594b–595b, 595f
- legibility, 598–599
- noticeability, 597–598
- presentation complexity, 600
- presentation timing, 600–601, 600b–601b
- visibility, 596–597, 597f

Feed-forward concept, 589–590, 601–626, 601f
 avoiding errors with, 623–624
 clarity, 602
 cognitive directness, 617–620, 617b–619b
 complete information, 620–621, 621b
 consistency of, 606–609, 606b–609b, 607f, 609f
 control complexity, 609, 609b
 data value formats treatment, 603–604
 distinguishability of choices, 604–606,
 605b–606b
 error recovery, 624
 example, 613b–614b, 625b–626b
 group-related elements, 610
 human memory limitations in, 615–617, 615b
 modes, 624–626
 precise wording, 602–603, 602b
 unrelated interaction elements, 610–614,
 611f–612f
 user/usage centeredness, 622–623, 622b–623b

Feedback to process, 484–485

Fidelity, prototyping, 349–351
 behavior, 350–351
 decoy user interface objects, 355
 design idea exploration in generative design,
 351–352
 dimensions of prototype fidelity, 350–351
 increasing levels of, 351–358
 low fidelity, 351
 low- to medium-fidelity wireframe prototype,
 357b–358b
 medium- to high-fidelity click-through prototypes,
 354–355
 medium- to high-fidelity prototypes, 356–358
 medium-fidelity click-through wireframe prototypes,
 354
 presentation, 350
 this feature not yet implemented, 355, 356f
 UX team, 357–358
 very low- fidelity wireframe, 351–352
 wireframe decks, 352
 wireframes for subsequent design walk- throughs,
 352–353, 353f

Final pilot testing, 409

Fitts, P. M., 21

Flow model, 131–132, 139–143, 230
 central importance, 139
 customer journey map, 143
 defined, 139
 for product/system, 142b–143b
 making of, 139–143
 non-UI software functionality, 140

Focus group, 93

Foley, J. D., 23

Formal summative evaluation, 371–372

Framing, 36

Frank, B., 134–135

Fun and usability, 12

Functional affordance, 556–557, 557f

Funnel model, agile UX method
 early-funnel activities, 65–67, 66b
 late- funnel activities, 63–65, 64f
 new model needed for, 62
 scope, 63
 speed and rigor in, 63, 64f

G

Gajendar, U., 220

Gear-shift pattern, 548b

Generative design, 284, 297
 activities, 225
 critiquing, 226f, 250–251
 ideation, 226f
 aircraft flight recorders, 243b–244b
 building blocks, 226–227
 catalysts, 227, 234–235
 creative role, 227
 informers, 227
 practice in, 248b–249b
 techniques, 227, 235–241
 immersion, 224–225
 iteration, 202–203, 204f
 preparing for, 224–225
 rules of engagement, 251–252
 sketching, 226f
 characteristics, 244–245
 physical mockups, 249–250

Generative design (*Continued*)

- practice in, 248b–249b
- synthesis, 225
- Gladwell, M., 235
- Gothelf, J., 178
- Graphic/visual designer, 74
- Graphical user interfaces (GUIs), 4, 23
- Green, T. R. G., 23
- Gross, M., 254
- Group-related UX problem descriptions, 469–470
- GUIs. *See* Graphical user interfaces (GUIs)
- Gulf of evaluation, 570
- Gulf of execution, 569–570

H

- Hallvard, T., 298
- Harrison, S., 40
- Hassenzahl, M., 11
- HCI. *See* Human-computer interaction (HCI)
- Henderson, A., 265
- Heuristic evaluation, 451–456, 454f
 - documenting UX problems, 454
 - heuristics, 451–452
 - limitations, 455–456
 - procedure, 452–453
 - variations abound, 454–455
- Hierarchical indented lists, 119
- Hierarchical lists, 33–34, 33f
- Hierarchical task inventory (HTI), 179, 182
 - for product/system, 157b
 - hierarchical relationship, 155–156
 - task structure model, 154–157
 - vs.* functions, 155
 - temporal sequencing, 156f
 - user work role, 157
- High-fidelity physical prototyping, 361–362
- Holtzblatt, K., 139, 159–160, 196, 211
- Horizontal prototypes, 340, 342
- Houben, S., 277
- Human factors engineering, 21
- Human memory
 - long-term memory, 587–588
 - muscle memory, 586–587, 587b
- Human memory (*Continued*)
 - other kinds of, 586–587
 - sensory memory, 586
 - short-term/working memory, 584–586
- Human working memory loads, 626–627
- Human-artifact interaction, 8f
- Human-centered design (HCD), 210–211
- Human-computer interaction (HCI), 3, 105, 253
 - cognitive science, 21–22
 - contextual studies into, 106
 - designers, 361
 - distinction from, 9
 - hardware and software foundations, 22, 22f
 - history, 20–24
 - human factors, 21–22
 - psychology, 21–22
 - research, 365
- Hybrid model, 171

I

- Ideal mental model, 256–258, 265
- Ideation, 226f, 247f
 - aircraft flight recorders, 243b–244b
 - at IDEO, 242b–243b
 - framing, 235–237
 - generative design
 - aircraft flight recorders, 243b–244b
 - building blocks, 226–227
 - catalysts, 227, 234–235
 - creative role, 227
 - informers, 227
 - practice in, 248b–249b
 - techniques, 227, 235–241
 - incubation, 238
 - magic wand technique, 237
 - participatory design, 238
 - reframing, 235–237
 - rules of engagement, 251–252
 - sketching, 244–245
- Immersion, 35, 147b
- Impulse buying, 294
- Incentives, 402
- Incubation ideation technique, 238

- Individual differences, 378
- Informal summative evaluation, 370–371
- Information, 150–151, 298–301
- Information architecture, 102, 150–151, 150b–151b, 275–276, 298–301
 - model, 131, 150–151, 150b–151b, 231–232
 - pervasive, 276
- Informed consent, 406–407
- Informers, ideation, 227
- Initial user performance, 387b
- Interaction
 - concept, expanding, 4–5
 - modes of, 5
 - new interaction artifacts, 4–5
 - emotional impact, 11
 - evolution of, 4
 - fun, 12
 - techniques, 22–24
 - user experience
 - activity-based, 5–7
 - expanding concept, 4–5
 - results from, 8
 - system-spanning, 6
- Interaction artifact model, 4–5
 - modes of, 5
- Interaction channels, 276–277
- Interaction cycle, 588–589, 589f
 - cooperative user-system task performance, 576–578
 - intention shifts, 577–578
 - path variations, 576–577, 577f
 - primary tasks, 576
 - secondary tasks, 577–578
 - stacking, 577–578, 579f
 - defined, 567
 - gulfs between user and system, 569–570
 - gulf of evaluation, 570
 - gulf of execution, 569–570
- Norman’s model to, 570–572
 - outcomes and system response, 571–572
 - partitioning, 571, 571f
- Norman’s stages-of-action model, 568–572, 568f
- theory-based conceptual framework, 567
- UX design issues, 572–576
- Interaction cycle (*Continued*)
 - assessment, 574–576, 575b–576b
 - outcomes, 574
 - physical actions, 573–574
 - planning, 572
 - translation, 572–573
- Interaction design
 - conceptual design, 297–298
 - creation of, 297–301
 - defined, 295–296
 - design system, 321–322
 - adopt and maintain, 320–322
 - defined, 320–321
 - designers reasons, 321
 - device types, 296–297
 - ecology roles, 297
 - information architecture, 298
 - intermediate interaction design, 307–316
 - client and users feedback, 314–316, 315f
 - final product experience presentation fidelity, 316
 - national park campground reservation system, 310–313, 311f–313f
 - National Park Website, 308–316
 - product/system, user persona for, 310b
 - patterns, 298
 - production, 316–320, 317f–319f
 - segue, 322
 - storyboards, 301–304
 - supporting tasks, 295–296
 - wireframes
 - defined, 305
 - design elements, 307
 - path, 305–307
- Interaction level, pyramid of needs, 374
- Interaction technique, 201, 299b
 - conceptual design, 299f, 299b, 301b
- Interactive graphics, 22–24
- Intermediate design
 - emotional impact design
 - audio characteristics, 331–332
 - motion styles and physics, 329–331, 330f
 - social and psychological aspects, 332
 - tone of language, 331

Intermediate design (*Continued*)
 visual language and vocabulary, 328–329, 329b
 iteration, 204–206, 204f
 Intertwining lifecycle activities
 examples of, 69–70
 lifecycle activity timing, 70–71
 Intrinsic methods, 373
 iPad, 297–298
 iPhone, 542f
 Ishii, H., 254
 Iteration design
 conceptual, 204
 detailed, 206
 generative, 202–203
 intermediate, 204–206

J

Job-related activities, 86
 Johnson, J., 265
 Jones, B. D., 21
 Just-in-time towel dispenser message,
 553–554, 554f

K

Kellogg, W. A., 269
 Keynote, 12
 Kolko, J., 37, 184
 Krippendorff, K., 235

L

Lab/equipment, 403, 404f
 Large and complex systems, 46
 Late-funnel data gathering, 95
 Late-funnel activities, 63–65, 64f
 Late-funnel mini-UX process checklist, 519t
 Lazy hand-waving, 497
 Leverage design patterns, 298
 Lifecycle activities, 28–29
 concepts, 28
 design solutions, 29
 evaluation, 29
 mapping project parameters, 40–41, 41f
 prototype candidates, 29

Lifecycle activities (*Continued*)
 understand needs, 28
 UX design lifecycle process, 28f
 Lifecycle process choice, UX, 27, 28f
 Lifecycle process, agile UX method, 58–61
 agility without chunking, 60
 chunking, 59–60, 59f
 reasons for, 61
 scope and elicit quick feedback, 59–60
 software engineering, 60–61
 spectrum, bookends on, 59
 user-facing prototypes, 60–61
 users, interest in, 61
 Lifecycle sub-activities, 29–30
 Line of affordability, 479–481, 480t, 483
 Liu, M., 11
 Local prototypes, 341–343
 Lockwood, L. A. D., 155, 165–166
 Long-term memory, 587–588
 recognition *vs.* recall, 587–588
 shortcuts, 588
 Lowgren, J., 360–361
 Lubow, A., 220b–221b

M

Mack, R. L., 269
 Magic wand ideation technique, 237
 Mail icon, 535b
 Manage finances, 156b
 Managing change challenges, agile UX method, 55
 lifecycle process, 56
 timely feedback, 56
 waterfall model, software system development, 56–58, 57f
 Managing customer stories and development tasks, 509
 Manual dexterity, 546
 Market differentiator, 323
 Marketing, emotional impact, 14–15
 Mason, J. G., 252
 Meaningfulness, 16, 102–103
 Meaningfulness, data collection, 427–429
 diary-based self-reporting by users, 428–429
 evaluation requires long-term studies, 428
 self-reporting, importance of, 428

- Melguizo, M. d. C. P., 258
 - Mental models, 263–264, 265b
 - conceptual design
 - as connection, 256–265
 - as mapping, 259–262
 - design patterns, 269
 - emotional impacts, 268
 - interaction, 267–268
 - metaphors, 269–271
 - work practice ecology, 266–267
 - defined, 255–256
 - designer's, 257f, 258
 - ideal, 257–258
 - user's, 258–259
 - Wheel UX lifecycle, 255–256, 256f
 - Menu planning application, 159b
 - Metaphors, 297–298, 326
 - conceptual design, 269–271
 - Middleburg University Ticket Transaction
 - Service (MUTTS), 75–76, 115, 134b, 181b, 391b
 - flow model, 140b, 141f
 - hierarchical task inventory diagram, 156f
 - ticket seller role, 158f
 - information architecture modeling, 150b–151b
 - physical work environment model, 148b, 149f
 - simple early flow model, 140b, 141f
 - social model, 169b–171b
 - task sequence model
 - branching and looping, 164b, 165f
 - rudimentary, 161b
 - step-by-step task interaction model, 161b–163b
 - ticket buying, 140b
 - user class definition, 136b
 - work role identification, 134b
 - Mind-mapping, 34
 - Minimum viable product (MVP), 59–60
 - Modeling, 37–38
 - Modes of interaction, 5
 - Mood boards, 327–328, 327f
 - Moore, N. C., 220
 - Motion styles and physics, 329–331, 330f
 - Muller, M. J., 6b, 238
 - Multiple problem solutions, 483
 - Myers, B. A., 23
- ## N
- National Bureau of Standards, 20
 - Newman, W. M., 23
 - Nielsen, J., 18
 - Nilsson, P., 244
 - Non-disclosure agreement (NDA), 408
 - Norman, D. A., 5, 10, 17, 210, 258–259, 265, 277, 285
 - model, 570–572
 - outcomes and system response, 571–572
 - partitioning, 571, 571f
 - stages-of-action model, 568–572, 568f
 - Note taking, 32
- ## O
- Objective data collection, 415
 - Observation, 31–32
 - Olsen, D. R., Jr., 23
 - Olson, J. R., 258
 - OmniGraffle, 12
 - Onion layer effect, 485–486, 485b–486b
 - Ottersten, I., 244
- ## P
- Pérez-Quinones, M. A., 277
 - Participants preliminaries, 429
 - Participation/buy-in, 399
 - Participatory design, 238
 - embodied and tangible interaction, 253–254
 - history related to, 253
 - Payoff methods, 373
 - Persona model, 228–230, 229f
 - Personal computing, 23
 - Pervasive information architecture, 276
 - Physical actions, UX design guidelines
 - avoiding physical overshoot errors, 639–640, 640b
 - awkwardness, 642
 - fatigue, 642
 - Fitts law, 637–638
 - help user in, 636–642, 636f
 - manual dexterity, 637–638

- Physical actions, UX design guidelines (*Continued*)
 - physical disabilities, 642
 - physical movements inertia, 640–641, 641f, 641b
 - sensing objects of, 634–636
 - to manipulate, 635–636
- Physical affordances
 - cart button as, 541b
 - constraining physical actions, 547–548
 - definition of, 540–541
 - design issues, 548–549
 - experienced/power users, 541
 - Fitts law, 546
 - gear-shift pattern, 548b
 - iPhone, 542f
 - manual dexterity, 546
 - opening doors, 542–543, 543f
 - physical devices, 543
 - cognitive, 543
 - emotional, 543, 544f
 - physical overshoot, 546–548, 547f, 547b
 - physicality, 544–546, 545b
 - user, 541–542, 541f
- Physical devices, 543
 - cognitive affordance, 543
 - emotional affordance, 543, 544f
- Physical interactivity, physical mockups for, 358–360, 359f
- Physical mockup, as embodied sketches, 249–250, 249f
- Physical models, 230
- Physical overshoot, 546–548, 547f, 547b
- Physical work environment model, 102, 131, 148b
 - environmental factor, 148
 - hardware design, 147
- Physicality, 544–546, 545b
- PICTIVE³ approach, 253
- Planning, 507–509, 507f
- Planning, UX design guidelines, 590–592, 590f
 - avoiding transaction completion slips, 592, 592b
 - clear system task model, 590–591, 591b
 - efficient task paths, 591–592, 591b–592b
- Pogue, D., 14, 14b
- Post-deployment field testing, 437
- Powerlines, system-spanning activity, 6b, 7f
- Priority rankings, 477–479, 479f
- Priority ratio, 475–477, 476b, 477t
- Process-related exercises, 79–80
 - exercises, 79
 - product concept statement for, 80b
 - product/system, 79–80
- Product concept statement
 - defined, 78
 - nature of, 77
- Product owner/manager, 74
- Product/system concept statement, 77–80
- Professional *vs.* personal goal, 172–173
- Project commission, UX team roles, 72–73
 - graphic/visual designer, 74
 - product owner/manager, 74
 - project/program manager, 75
 - software engineer, 75
 - UX analyst/evaluator, 74
 - UX designer, 74
 - UX researcher, 73
- Project resources influence, 51
- Project/program manager, 75
- Prototype evaluating, 429–430
- Prototype executor, 400
- Prototyping, 337–338, 338f
 - advantages, 339
 - depth and breadth, 340f
 - examples of kinds, 342–343
 - horizontal prototypes, 340, 342
 - local prototypes, 341–343
 - T prototypes, 341–343
 - vertical prototypes, 340–342
 - dilemma and solution, 338
 - envisioning, 38
 - fidelity, 349–351
 - behavior, 350–351
 - decoy user interface objects, 355
 - design idea exploration in generative design, 351–352
 - dimensions of prototype fidelity, 350–351
 - increasing levels of, 351–358
 - low fidelity, 351
 - low- to medium-fidelity wireframe prototype, 357b–358b

Prototyping (*Continued*)

- medium- to high-fidelity click-through prototypes, 354–355
- medium- to high-fidelity prototypes, 356–358
- medium-fidelity click-through wireframe prototypes, 354
- presentation, 350
- this feature not yet implemented, 355, 356f
- UX team, 357–358
- very low-fidelity wireframe, 351–352
- wireframe decks, 352
- wireframes for subsequent design walk-throughs, 352–353, 353f
- intertwines with, 338
- Scandinavian origins of, 362
- segue, 362
- specialized prototypes, 358–362
 - animated prototypes, 360–361
 - experience prototyping, 361–362
 - high-fidelity physical prototyping, 361–362
 - physical interactivity, physical mockups for, 358–360, 359f
- universality of, 339
- wireframe, 343–349, 344f
 - creating Each state (box), 346–347
 - decks, 347
 - design, 345
 - library of interaction object templates, 347–349, 348f
 - represent flow and navigation, 345–346, 345b
 - software tools, 362
- Purity, 44
- Pyla, P. S., 277

Q

Qualitative data, 367–369

- analysis
 - preparation steps, 466
 - steps, 466–471, 467f
- collection
 - co-discovery think-aloud techniques, 426
 - critical incident technique, 422–424
 - user think-aloud data collection techniques, 425–426

Qualitative data (*Continued*)

- collectors, 400
- overview, 465
- Quantitative data, 367, 461–465
 - big decision, 464–465
 - collection
 - counting user errors, 415
 - objective data collection, 415
 - questionnaires, 416–421
 - slip is not a user error, 415
 - subjective data collection, 416–421
 - timing task performance, 415
 - collectors, 400
 - convergence, 465
 - line up your quantitative ducks, 463–464
 - meet target column, 463–464
 - observed results, 463
 - simple descriptive statistics, 462–463
 - subjective quantitative questionnaire data, 463
- Quasi-empirical UX evaluation, 433–434
- Questionnaire for User Interface Satisfaction (QUIS), 388, 417, 418t
- Questionnaires, 408–409, 416–421
 - advantages of, 416
 - customer satisfaction (CSAT) studies, 421
 - empirical data, 416
 - general-purpose usability questionnaires, 419
 - multi-media system evaluation questionnaires, 420
 - semantic differential scales, 416
 - system usability scale, 418–419
 - user interface satisfaction, 417
 - web evaluation questionnaires, 420

R

- Range of rigor, 374–375
- Rapid cost-importance analysis, 484
- Rapid empirical UX evaluation methods, 433–436
 - introduction to, 433
 - quasi-empirical UX evaluation, 433–434
 - rapid iterative testing and evaluation (RITE), 434–436
 - variations, 436
- Rapid iterative testing and evaluation (RITE), 434–436

Rapid UX evaluation (discount) methods, 378

Reasoning and deduction, 36

Recruit participants, 401–403

- budget and schedule for, 401
- codiscovery evaluation, 402–403
- identify right kinds of participants, 401
- incentives and remuneration, 402
- recruiting methods and screening, 402
- right number of participants, 402
- subsequent iterations, select participants for, 403

Recruit user participants, 95–96

Recruiting methods and screening, 402

Reframing, 36

Regression testing, 522

Reliability, 26

Reliability of formative UX evaluation methods, 377–380

Remote UX evaluation, 437–438

Remuneration, 402

Reporting results, user experience (UX) evaluation, 487

- audience considerations, 490–493
- caution about reporting summative results, 489–490
- common industry format, 488–489
- cross-functional team, 489
- guidelines, 493–497
- mechanics of, 496–497
- participant anonymity, 488
- personal presence, 490
- problem reporting, 493–496
- quality communication, 487–488
- tone and style, 497

Resistance to innovation, 46

Resmini, A., 232, 277, 289, 293

Ries, E., 59–60

Rigor, design situations, 43–51

- complexity influences, 44–48
- development stages, 50
- domain familiarity influences, 49
- project resources influence, 51
- risk aversion influences, 49–50
- UX team experience, 51

Risk aversion influences, 49–50

- risk due to data loss, 49–50
- risks associated with legal, safety, and compliance constraints, 50

Rosati, L., 232, 277, 289, 293

Royce, W. W., 57

Russell, D. M., 4

S

Scenario, 158–159

Scope of delivery, 178

SCRABBLE Flash Cube game, 240b–241b, 241f

Scriven, M., 376–377

Seiden, J., 178

Self-reporting technique questionnaires, 427

Self-reporting, importance of, 428

Self-sufficiency, 285–287

Sensory affordance

- auditory sensory issues, 555, 555b
- definition of, 549
- design of, 549
- haptic and tactile sensory issues, 556
- visual sensory affordance design issues, 550–554
 - black-on-black in stereo, 550–552, 552f
 - example, 550
 - just-in-time towel dispenser message, 553–554, 554f

Sequence models, 230–231

Served users, 133–134

Service design, 18

Service experience, 18–19

Session parameters, 404–406, 405f

Shadowing user, 143

Shared conventions, 535–536

Shared cultural conventions, 528–531

Shih, Y. -H. X., 11

Short-term/working memory, 584–586

- chunking, 584–585, 584b–585b
- cognitive load, 585–586
- stacking, 585

Simple descriptive statistics, 462–463

Siri, 295–296

Sketching, 32, 226f

- embodied cognition, 245

- Sketching (*Continued*)
 - freehand gestural, 247f
 - generative design
 - characteristics, 244–245
 - physical mockups, 249–250
 - practice in, 248b–249b
 - rules of engagement, 251–252
 - Slanty design, 287–288
 - Sloppy terminology, 497
 - Small upfront analysis and design, 516–517
 - Smart shopping application, 289–294
 - high-level issues, 289–290
 - key parts of design, 290
 - Smartwatches, 4, 6
 - Smith, K. C. M., 235
 - Social model, 168–171, 232–234, 233b
 - active entity, 169
 - concerns, 169–171
 - for product/system, 171b
 - influences, 169
 - issues with, 169–171
 - shared workplace culture, 168
 - simplified approach, 168–169
 - Social/psychological aspects, 332
 - Software engineering (SE), 39, 60–61, 177
 - Soviet psychological activity theory, 105
 - Special Interest Group for Social and Behavioral Science Computing (SIGSOC), 20–21
 - Specialized UX evaluation methods, 437–441
 - A/B comparison testing, 438–441
 - alpha and beta testing and field surveys, 437
 - automatic UX evaluation, 438
 - post-deployment field testing, 437
 - remote UX evaluation, 437–438
 - Sprints, 517–519, 518f
 - acceptance test creation, 510
 - acceptance testing and deployment, 511
 - code testing, 510
 - implementation coding, 510
 - regression testing, 511
 - unit code test creation, 510
 - Stacking, 585
 - Story-based planning, 508, 508f
 - Storyboards, 279–283
 - Storytelling, 38
 - Streitz, N. A., 4
 - Structural consistency, 663–665
 - Subject-matter expert (SME), 89, 93
 - Subjective data collection, 416–421
 - Subjective quantitative questionnaire data, 463
 - Subjective UX data, 369
 - Subjective UX evaluation data, 256
 - Subsequent iterations, select participants for, 403
 - Summative UX evaluation, 370
 - Supplemental data sources, 93
 - Suri, J. F., 361
 - Surveys, 408–409
 - Sutherland, I. E., 23
 - Synthesis, 37, 225
 - System complexity space, 175–176
 - System functionality, 643
 - System response time, 644
 - System-spanning interaction, 6, 7f
- T**
- T prototypes, 341–343
 - Tangible interaction, 240–241, 254
 - Target level, 391–394, 392t–393t, 392b
 - Task completion errors, 538, 538f
 - Task model, 132
 - Task sequence model, 131, 161b
 - beyond linear model, 163–164
 - branching, 163–164, 164f
 - components
 - step goals, 159
 - step trigger, 160
 - task barrier, 160
 - user information and other needs, 160
 - defined, 158
 - essential use case, 165–167
 - for product/system, 159b
 - looping, 163–164, 164f
 - menu planning application, 159b
 - step-by-step description, 159–163, 161b, 164f
 - usage scenario, 159b

Task structure model, 131
 benefits, 154

Task thread continuity, 628–629, 628b–629b

Tasks *vs.* functions, 155

Tatar, D., 40

Team roles, 399–400
 facilitator, 400
 participation and buy-in, 399
 prototype executor, 400
 qualitative data collectors, 400
 quantitative data collectors, 400

Technical program manager (TPM). *See* Project/
 program manager

Theory-based conceptual framework, 567

Thermostat ecology, 258, 266, 268f

Thomas, J. C., 269

Ticket Kiosk System (TKS), 76, 111b–113b, 122b–123b,
 135, 470b
 elemental data note, 186b
 essential use case, 166b, 167t
 interaction
 conceptual design, 299f, 299b
 traditional concept for, 300–301, 300f
 physical hardware requirements, 185b
 product concept statement for, 78–79
 rationale, 77
 social model entities, 169b
 system support requirements, 186b
 user experience (UX)
 design requirements, 175, 177, 184
 evaluation, 385b
 user persona in, 136–138, 137b–138b
 user stories, 175–176, 183b

Tidwell, J., 298

Tie-breakers, 483

Timing task performance, 415

Tohidi, M., 244

Tone of language, 331

Top-down UX design approach
 abstract work activities, 215
 abstract work activity, 214–215
 characteristics, 219–220
 disruptive design, 220–221

Top-down UX design approach (*Continued*)
 in architecture, 220b–221b
 in architecture, 220b–221b
 work, 213
 work activity, 213–214
 work domain, 214
 work practice, 214
 work role, 213

Totality of effects, of interaction, 8

Tungare, M., 240, 277

Two agile workflows, synchronizing, 520–523
 communication during synchronization, 523
 continuous delivery, 522
 dovetailed work activities, 520–521, 520f
 planning across iterations, 522–523
 regression testing, 522
 value of early delivery, 521–522

U

Ullmer, B., 240

Understand Needs lifecycle activity, 28

Unmeasured tasks, 401

Usability, 9–10, 377
 dancing bear software, 16
 efficiency and effectiveness, 11
 engineering, 16
 initial tolerance of, UX design, 16

Usage research data modeling, 129–131
 analysis, 131
 artifact model
 defined, 144
 flow model, 140b
 central importance, 139
 customer journey map, 143
 defined, 139
 for product/system, 142b–143b
 making of, 139–143
 hybrid model, 171
 information architecture model, 150–151
 kinds of model, 131
 physical work environment model
 environmental factor, 148
 hardware design, 147

Usage research data modeling (*Continued*)

- social model
 - active entity, 169
 - concerns, 169–171
 - for product/system, 171b
 - influences, 169
 - issues with, 169–171
 - shared workplace culture, 168
 - simplified approach, 168–169
- task sequence model, 161b, 167b
 - beyond linear model, 163–164
 - defined, 158
 - essential use case, 165–167
 - for product/system, 159b
 - menu planning application, 159b
 - step-by-step description, 165–166
- the wheel, 130f
- user persona model
 - creation, 138
 - defined, 137
 - early sketch, 138b
 - Ticket Kiosk System, 138b
- work role model, 134b
 - defined, 132–133
 - for product/system, 134b
 - identification, 134b
 - mediated work role, 133–134
 - subroles, 133
 - user class, 134–136
- Usage scenario, 159, 159b
- Use think-aloud technique, 427
- Useful defaults, 629–631, 630b–631b
- Usefulness, 10
- User class definition, 136b
 - defined, 134–136
 - for product/system, 136b
 - knowledge-based characteristics, 135
 - physiological characteristics, 135–136
 - skills-based characteristics, 135
 - work role modeling, 136
- User classes, 382–383

User experience (UX)

- analyst/evaluator, 74
- appropriating methods and techniques, 39–41
- basic process components
 - design lifecycle process, 26
 - hierarchy of terms, 27
 - lifecycle activities, 28–29
 - lifecycle sub-activities, 29–30
- branding, 14–15
- business case, 19–20
- characteristics, 7
- choosing methods and techniques, 40
- components, 10f
 - emotional impact, 10–15
 - meaningfulness, 16
 - usability, 9–10
 - usefulness, 10
- computing, 4
- context and ecology, 9
- cost justification, 20
- defined, 7
- design guidelines
 - examples, 582
- feed-forward concept, 589–590, 601–626, 601f
 - avoiding errors with, 623–624
 - clarity, 602
 - cognitive directness, 617–620, 617b–619b
 - complete information, 620–621, 621b
 - consistency of feed-forward, 606–609, 606b–609b, 607f, 609f
 - control complexity, 609, 609b
 - data value formats treatment, 603–604
 - distinguishability of choices, 604–606, 605b–606b
 - example, 613b–614b, 625b–626b
 - feed-forward for error recovery, 624
 - feed-forward for modes, 624–626
 - group-related elements, 610
 - human memory limitations in feed-forward, 615–617, 615b
 - precise wording, 602–603, 602b
 - unrelated interaction elements, 610–614, 611f–612f
- user/usage centeredness, 622–623, 622b–623b

User experience (UX) (*Continued*)

- history, 674–675
- human memory
 - long-term memory, 587–588
 - muscle memory, 586–587, 587b
 - other kinds of, 586–587
 - sensory memory, 586
 - short-term/working memory, 584–586
- interaction cycle as organizational structure, 588–589, 589f
- interpreting, 582–583
- outcomes, 643–646
 - automation issues, 644–646, 644b–646b
 - system functionality, 643
 - system response time, 644
- overall
 - anthropomorphism, 666–669
 - consistency, 662–665, 662b–663b
 - helpful help, 674
 - humor, 666
 - reducing friction, 665–666
 - simplicity, 661–662, 662b
 - sound and color, use of, 669–673, 670b, 671f, 672b
 - structural consistency, 663–665
 - tone and psychological impact, 669
 - user differences accommodation, 673
 - user preferences, 673
 - users in control, 659–661, 660b
- planning, 590–592, 590f
 - avoiding transaction completion slips, 592, 592b
 - clear system task model, 590–591, 591b
 - efficient task paths, 591–592, 591b–592b
- scope and universality, 581–582
- task structure, 626–633
 - design efficiency, 627–628
 - design flexibility, 627
 - human working memory loads, 626–627
 - not undoing user work, 632
 - retaining essential user work context, 629, 629b
 - retaining state information, 632–633, 632b–633b
 - task thread continuity, 628–629, 628b–629b
 - useful defaults, 629–631, 630b–631b

User experience (UX) (*Continued*)

- translation, 593–633
 - color, 600
 - discernibility, 598
 - distinguishability, 600
 - existence of feed-forward, 593–595, 593f, 594b–595b, 595f
 - feed-forward legibility, 598–599
 - feed-forward noticeability, 597–598
 - feed-forward presentation complexity, 600
 - feed-forward presentation timing, 600–601, 600b–601b
 - feed-forward visibility, 596–597, 597f
 - presentation of, 596–601, 596f
- designer, 74
- ecology, 5
- engagement, 11
- enticement, 11
- fuss about usability, 19
- history, 20–24
- inspection
 - broad range of, 450
 - co-discovery approach, 456–457
 - defined, 449
 - heuristic evaluation, 451–456
 - insight and experience, 456
 - inspectors needed for, 450–451
 - kind of expertise, 450
 - practical approach to, 456–457
 - rich and comprehensive usage-oriented view, 457
 - tasks and design driven, 457
- interaction, 4
 - activity-based, 5–7
 - expanding concept, 4–5
 - results from, 8
 - system-spanning, 6
- intertwining lifecycle activities
 - examples of, 69–70
 - timing, 70–71
- lifecycle activities, 28–29
- lifecycle activity, 139
- lifecycle process choice, 39
- mapping project parameters, 40–41, 41f

User experience (UX) (*Continued*)

- metrics, 389–391, 390t
 - targets, project context for, 381–382
- process
 - consistency, 26
 - guiding structure, 26
 - learning, 26
 - need for, 26
 - reliability, 26
 - shared conception, 26
- project commission, 72–73
 - graphic/visual designer, 74
 - product owner/manager, 74
 - project/program manager, 75
 - software engineer, 75
 - UX analyst/evaluator, 74
 - UX designer, 74
 - UX researcher, 73
- researcher, 73
- rise of, 17
- service experience, 18–19
- software engineering lifecycle, 25
- target tables, 382
- technique toolbox, 104
- techniques, 30–39
- totality of effects, 8

User experience (UX) design

- and costs of, 17–18
- architecture, 197–198
- attractive designs, 11
- brand, 14–15
- checklist, 559–562
- coolness and wow design, 14
- corporate culture, 14–15
- design informing model, 129–130
- guidelines, assessment
 - feedback, 646
 - completeness of, 651, 651f, 651b
 - consistency of, 654
 - content and meaning of, 649–654, 649b
 - feedback noticeability, 648–649
 - feedback timing, 649, 649b

User experience (UX) design (*Continued*)

- information displays assessment, 655–659, 656b–657b
- medium, 649
- tone of, 652
- usage centeredness, 652–654
- user control, 654
- visual bandwidth for information display, 657–659, 657b–659b, 658f–659f
- system feedback assessment, 647–648
 - progress indicators, 648
- system response, 647
- guidelines, physical actions
 - avoiding physical overshoot errors, 639–640, 640b
 - awkwardness, 642
 - fatigue, 642
 - Fitts law, 637–638
 - help user in, 636–642, 636f
 - manual dexterity, 637–638
 - physical disabilities, 642
 - physical movements inertia, 640–641, 641f, 641b
 - sensing objects during manipulation, 635–636
 - sensing objects of, 634–636
 - sensing objects to manipulate, 635
- importance of, 17–18
- interdisciplinary nature, 198
- issues, 572–576
 - assessment, 574–576, 575b–576b
 - outcomes, 574
 - physical actions, 573–574
 - planning, 572
 - translation, 572–573
- lifecycle process, 27, 28f, 176f
- marketing, 14–15
- methods, 30
- pyramid of human needs, 201f
- requirements, 175–176
 - compose requirements statements, 187
 - generic structure, 185f
 - software and functional implications, 180b–181b
 - team selection, 188
 - vs.* prototypes, 189
- situations, 40

User experience (UX) design (*Continued*)

- studio, 139–140, 173
 - dedicated workspace, 71–72
 - reason for, 71
 - Virginia Tech Industrial Design Studio, The Kiva, 72
 - task barrier, 160
 - task models, 154
 - techniques as life skills, 30–31
 - abstraction, 35–36
 - brainstorming, 38
 - framing and reframing, 36
 - immersion, 35
 - modeling, 37–38
 - note taking, 32
 - observation, 31–32
 - prototyping for envisioning, 38
 - reasoning and deduction, 36
 - sketching, 32
 - storytelling, 38
 - synthesis, 37
 - universality, 196
 - user stories, 175–183
 - cleaning up, 178–179
 - defined, 178–179
 - extrapolation requirements, 181
 - team selection, 180b
 - ticket buyer work role, 182f
 - validation of, 188
 - wants, design, 175–176
- User experience (UX) evaluation, 365–366, 366f, 381–382
- adapt and appropriate methods, 374–376
 - range of rigor, 374–375
 - audience considerations, 490–493
 - customers/clients, 492–493, 492b–493b
 - inform/influence management, 491
 - primary audience, 490–491
 - re-design, 491–492
 - baseline level, 391
 - empirical and analytic evaluation, 376–377
 - formal summative evaluation, 371–372
 - A/B testing, 371
 - CSAT studies, 371–372

User experience (UX) evaluation (*Continued*)

- formative UX evaluation, 370
 - analytic formative evaluation, 372–373
 - ecological level of the pyramid of needs, 373–374
 - emotional level of the pyramid of needs, 374
 - empirical formative UX evaluation, 372
 - payoff and intrinsic methods, 373
 - within the interaction level of the pyramid of needs, 374
 - within the pyramid of human needs, 373, 373f
- goals, 383–385, 384b
- goals, metrics, and targets, 382f
- informal summative evaluation, 370–371
- measurability of user experience parameters, 377
- measures, 385
 - Ticket Kiosk System, 385b
- measuring instruments, 385–389
 - benchmark tasks, 386–388
 - crafting benchmark task descriptions, 387–388
 - user satisfaction questionnaires, 388–389
- mechanics of report, 496–497
- objective UX data, 369
- observed results, 394–395, 394b–395b
- problem reporting, 493–496
 - content, 494–495
 - ecological and emotional layers of the pyramid, 496
 - foundation, 494
 - prioritize problems, 495–496
- qualitative data, 367–369
- quantitative data, 367
- reliability of formative UX evaluation methods, 377–380
- report guidelines, 493–497
- report tone and style, 497
- reporting results, 487
 - caution about reporting summative results, 489–490
 - common industry format, 488–489
 - cross-functional team, 489
 - participant anonymity, 488
 - personal presence, 490
 - quality communication, 487–488

User experience (UX) evaluation (*Continued*)

- subjective UX data, 369
- summative UX evaluation, 370
- target level, 391–394, 392t–393t, 392b
- types of, 367–369
- user classes, 382–383, 383b
- user testing, 366–367
- UX metrics, 389–391, 390t
- UX metrics and targets, project context for, 381–382
- UX target tables, 382
- work role, 382–383, 383b

User experience (UX) research

- analysis tools, 471
- data analysis, 107–108
 - by composing work activity notes, 108–114
 - goal of, 107–108
 - organize, 118–119
 - synthesize, 115–118
 - work activity affinity diagram, 119–127
- data gathering, 83–85, 103–104
 - collecting user data, 96–104
 - data sources, 93–94
 - effective raw data notes, 104
 - efficiency, 104
 - gathering data for understanding work domains
 - and practices, 88–90
 - goals based on scope, 95
 - nature of, 90–92
 - recruit user participants, 95–96
 - roots of user research, 105–106
 - team, 95
 - usage of products and systems, 85–87
 - visit parameters, 94
 - work practices within work domain, 88

User Interface Management Systems (UIMS), 23

User persona model, 100, 131

- creation, 136–137
- data extraction, 137–138
- defined, 137
- early sketch, 138b
- Ticket Kiosk System, 137b–138b

User satisfaction questionnaires, 388–389

User stories, 175–183

- cleaning up, 178–179
- defined, 178–179
- extrapolation requirements, 181
- team selection, 180b
- ticket buyer work role, 182f
- validation of, 188

User surveys, 93

User tasks and appropriate survey instruments, effective range of

- benchmark task descriptions, 400
- exploratory free use tasks, 401
- other kinds of task descriptions, 401
- unmeasured tasks, 401
- user-defined tasks, 401

User testing, 366–367

User think-aloud data collection techniques, 425–426

User think-aloud qualitative data collection techniques, 368

User work role model, 131–136, 228

User's mental model, 256, 257f, 258–259

User-centered design (UCD), 210–211

User-created affordances, 562–564, 562b–564b, 563f, 565f

User-defined tasks, 401

User-facing prototypes, 60–61

V

Vague descriptions, 497

Value-sensitive designs, 324

Veer, G. C. v. d., 258

Vertical prototypes, 340–342

Virginia Tech Industrial Design Studio, The Kiva, 72

Visual language, 328–329, 329b

Visual sensory affordance design issues, 550–554

- black-on-black in stereo, 550–552, 552f
- example, 550
- just-in-time towel dispenser message, 553–554, 554f

Vocabulary, 328–329, 329b

W

Waterfall lifecycle process, 184

Waterfall method, contrasting with, 504–506

- Waterfall model, software system development, 56–58, 57f
- Wearable devices, 4
- Weiser, M., 4, 277
- Welie, M. V., 298
- White, E. A., 6b
- Wildman, D. M., 6b
- Winograd, T., 4
- Wireflow prototype, 167
- Wireframes, 167, 189
 - prototyping, 343–349, 344f
 - creating Each state (box), 346–347
 - decks, 347
 - design, 345
 - library of interaction object templates, 347–349, 348f
 - represent flow and navigation, 345–346, 345b
 - software tools, 362
- Wixon, D. R., 379
- Wood, L. E., 196
- Work activity, 86
 - instances, 214
 - note, 134b, 179b–180b, 185b
 - notes, elemental parts of, 117
- Work activity affinity diagram (WAAD), 118–121, 126–127, 127b
- Work domain, 87, 89
 - complexity, 45–48
- Work environment model, 131, 147–148
 - environmental factor, 148
 - hardware design, 147
- Work practice, 86–87, 142b–143b, 214
- Work role model, 87, 132
 - defined, 132–133
 - for product/system, 134b
 - hierarchical task inventory, 154
 - mediated work role, 133–134
 - subroles, 133
 - user class, 134–136
- Workplace democracy movement, 253